

The Emergence of Edge Computing

The paper provides an overview of Edge computing, along with its potential advantages and some challenges. The author defines edge computing as the placement of computing and storage nodes, or cloudlets, at the internet's edge, in close proximity to mobile devices or sensors. Edge computing promises to deliver highly responsive cloud services for mobile computing, scalability and privacy policy enforcements for IoT devices, and the ability to mask transient cloud outages. The paper attempts to answer three major questions; why edge computing has emerged, what new capabilities does it enable and where it is headed.

The author starts by providing the origin and the background and states that the roots of edge computing reach back to the 90's when CDNs emerged. Edge computing extends and generalizes the CDN concepts by leveraging the cloud computing infrastructure. Like CDNs, proximity to end user is crucial for edge computing, however, unlike CDNs, it has applications beyond content caching. A cloudlet can run arbitrary code, just like in cloud computing.

The paper then discusses the importance of proximity to the user. Physical proximity affects end to end latency and economically viable bandwidth. The author argues that the proximity with cloudlets helps in at least four ways. Firstly, it enables highly responsive cloud services as close proximity provides high bandwidth, low latency access to services located on the cloudlet. Secondly, it provides scalability via edge analytics. For instance, cumulative ingress bandwidth from a large number of IoT devices is considerably reduced if the raw data is analyzed on cloudlets and only the extracted information and metadata are transmitted to the cloud. Thirdly, cloudlets can help enforce the owner's privacy policy before sending the data to the cloud. Lastly, these cloudlets can help mask short term cloud service outages.

The author then provides specific examples for all of these scenarios and discusses the road ahead. The author mentions the risks of weaker cloudlet perimeter security and proposes development of tamper resistant enclosures and the lack of incentive for deploying cloudlets.

Although the paper presents edge computing as the future of computing, it leaves me skeptical due to two major reasons. Firstly, I believe installing edge computing devices could increase the overall surface area for cyber security attacks. The author does touch on this in the road ahead section but their suggestion of tamper resistant enclosures is not convincing enough to say the least. Software vulnerabilities may still be easier to tackle but hardware vulnerabilities could prove disastrous. For example, a vulnerability affecting all Intel processors from 6th to 11th generation was disclosed some time ago. Such vulnerabilities are hard to deal with as the manufacturers can neither recall all devices that are affected, nor fix them. Such vulnerabilities could also plague edge computing hardware. Another problem, in my opinion, is that wide scale adoption of edge computing devices could increase the overall network complexity in unforeseen ways. One positive aspect of edge computing, however, is that it has the potential to make GDPR and other similar compliances easier.